SECURE BANKING NETWORK

Project Report

A Cisco-based network security and connectivity simulation project

Project Type: Academic / Portfolio Project

Platform: Cisco Packet Tracer

Focus: Network design, segmentation, routing, switching and device security

Prepared for GitHub Portfolio

1. Abstract

The Secure Banking Network project demonstrates the design and configuration of a secure enterprise-style banking network in a simulated Cisco environment. The project focuses on providing reliable connectivity while applying basic security controls at the router and switch level. The topology represents a practical banking environment where different network segments can be separated, managed and protected according to their purpose.

This project is intended as a learning and portfolio project rather than a production banking architecture. It demonstrates networking concepts such as IP addressing, VLAN-based segmentation, routing, switching, device access security and basic network management.

2. Project Objectives

- Design a structured banking network topology using Cisco networking devices.
- Create logical network segmentation to separate different departments or services.
- Configure IP addressing and basic routing for communication between network segments.
- Apply passwords and access controls to routers and switches.
- Demonstrate secure management practices for network devices.
- Test end-to-end connectivity and troubleshoot common configuration issues.
- Document the project so that it can be reproduced and reviewed through GitHub.

3. Project Scope

The scope of this project is limited to a simulated network created for educational and portfolio purposes. It does not represent a real bank's infrastructure, customer data environment, payment system or production security architecture. The design concentrates on core networking and introductory security concepts that are useful for a fresher network engineer.

4. Network Architecture

The topology is organized around network devices such as routers and switches, with end devices connected to the appropriate network segments. The architecture is designed to show how a banking organization can logically separate users or services while maintaining controlled communication between required segments.

Layer / Component	Purpose
End Devices	Represent users, systems or banking workstations in the simulated environment.
Access Switches	Connect end devices and provide Layer 2 connectivity and segmentation.
Router / Layer 3 Device	Provides routing between different networks or VLANs where required.
Security Configuration	Protects device management access using passwords and appropriate access controls.

5. Technologies and Concepts Used

Technology / Concept	Application in Project
Cisco Packet Tracer	Network simulation and configuration testing.
IPv4 Addressing	Assigning logical addresses to network interfaces and end devices.
VLAN / Segmentation	Separating network users or services into logical broadcast domains.
Routing	Enabling communication between different IP networks.

Switching	Providing local Layer 2 connectivity.
Device Password Security	Restricting unauthorized access to network device configuration.
CLI Configuration	Configuring routers and switches using Cisco IOS commands.

6. Security Implementation

Device Access Protection

Router and switch access is protected with configured passwords so that only authorized users can enter privileged or configuration modes.

Logical Segmentation

Separate network segments reduce unnecessary broadcast traffic and help isolate users or services.

Controlled Routing

Inter-network communication is performed through Layer 3 routing rather than allowing unrestricted Layer 2 access.

Secure Management Practice

The project demonstrates the importance of protecting network-device administrative access and avoiding exposure of credentials in public repositories.

7. Configuration Workflow

1. Plan the topology and identify the required network segments.
2. Assign an IP addressing scheme to routers, switches and end devices.
3. Configure switch ports and VLANs where required.
4. Configure router interfaces and routing between required networks.
5. Configure device hostnames and administrative passwords.
6. Verify interface status and IP addressing.
7. Test connectivity using ping and other basic troubleshooting commands.
8. Document the final topology, configurations and verification results.

8. Testing and Verification

After configuration, the simulated network should be tested systematically. The following checks can be used to validate the project:

- Verify that required router and switch interfaces are up.
- Use IP configuration checks to confirm correct addressing.
- Use ping tests between devices in the same network.
- Use ping tests between permitted different networks.
- Verify VLAN membership and trunk configuration where applicable.
- Check routing information and confirm that required routes are present.
- Attempt device management access to verify password protection.

9. Expected Results

The completed simulation should provide stable connectivity between the intended network segments while maintaining logical separation and protected device administration. The project should also demonstrate that the network can be configured, verified and troubleshot using Cisco IOS commands.

10. Skills Demonstrated

- Network topology design
- IPv4 addressing and subnetting
- Cisco IOS CLI
- Switch configuration
- Router configuration
- VLAN and network segmentation concepts
- Basic routing
- Network troubleshooting
- Device access security
- Technical documentation and GitHub project presentation

11. Limitations

This is a simulated portfolio project and should not be interpreted as a complete real-world banking security architecture. A production banking network would require many additional controls, including high availability, advanced firewalls, intrusion detection and prevention, centralized authentication, encryption, monitoring, logging, backup systems, compliance controls and extensive security testing.

12. Future Enhancements

- Add firewall zones and access-control policies.
- Introduce centralized authentication such as AAA.
- Add SSH-based secure device management.
- Implement DHCP, DNS and centralized services.
- Add redundancy and high-availability design.
- Add centralized logging and network monitoring.
- Expand the topology with additional branches or data-center services.
- Create detailed security policies and access-control rules.

13. Conclusion

The Secure Banking Network project provides a practical demonstration of fundamental networking and security concepts using Cisco Packet Tracer. It shows how a structured network can be planned, segmented, configured, secured and verified. As a portfolio project, it demonstrates hands-on knowledge of switching, routing, IP addressing, network segmentation, device security and troubleshooting, making it suitable for showcasing foundational skills for an entry-level network engineering role.